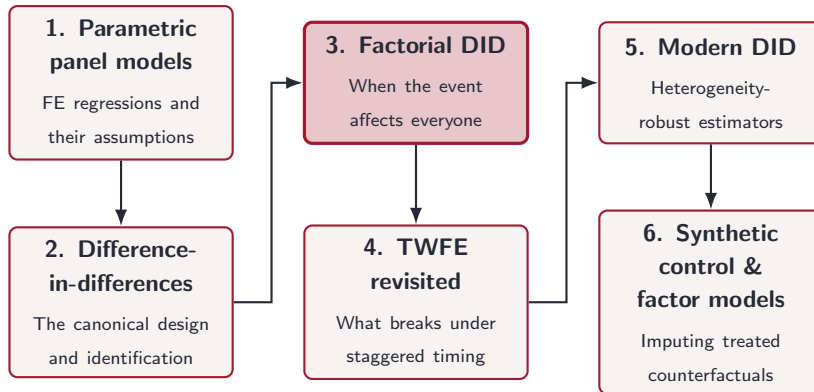


Panel Methods (3): Factorial Difference-in-Differences

Lectures on Causal Panel Analysis

Yiqing Xu
Stanford University

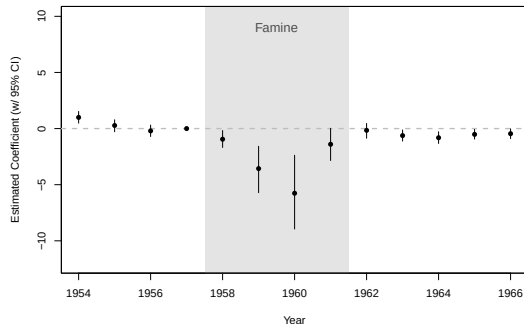
September 2026



- Lecture 2 covered the canonical DID design: a treated group, an untreated comparison group, and parallel trends
- But many events leave **no one untreated**: a famine, a financial crisis, a pandemic, a nationwide reform
- Applied work still runs “DID” in these settings, comparing units that differ in a **pre-event characteristic**
- This lecture: what does that comparison identify, and under which assumptions? (Xu, Zhao, and Ding 2026, *JASA*)
- Lecture 4 then returns to TWFE regressions and what breaks under staggered timing

Motivation: How Social Capital Saved Lives

- Cao, Xu, and Zhang (2022): “How social capital saved lives during China’s Great Famine”
- Data structure:
 - A baseline factor G : time-invariant measure of social capital
 - Event time: {Pre-Famine, Famine, Post-Famine}
- Estimation: difference-in-differences (DID), or equivalently, two-way fixed effects (TWFE)
- Interpretation:
 - **Descriptively**: “the rise in the mortality rate during the famine years is significantly smaller in counties with higher social capital”
 - **Causally**: “we interpret these differences as the effects of social capital on famine relief”



Estimates from TWFE with a dynamic specification.

How Do Immigrants Respond to Discrimination? The Case of Germans in the US During World War I

Vasiliki Fouka (2019). *American Political Science Review* 113 (2).

German immigrants (the baseline factor) meet the wave of anti-German hostility that came with the war (the event); outcomes are naming choices and petitions for naturalization.

Discrimination $\times$ WWI

Source: Fouka (2019).

Devotion and Development: Religiosity, Education, and Economic Progress in Nineteenth-Century France

Mara P. Squicciarini (2020). *American Economic Review* 110 (11).

More and less religious districts (the baseline factor) meet the Second Industrial Revolution's demand for technical schooling (the event); outcomes are education and industrial development.

Catholicism × Industrial Revolution

Source: Squicciarini (2020).

From Powerholders to Stakeholders: State-Building with Elite Compensation in Early Medieval China

Joy Chen, Erik H. Wang, and Xiaoming Zhang (2025). *American Journal of Political Science* 69 (2).

Localities with and without entrenched powerholders meet a state-building reform that reached everyone.

Elite strongholds $\times$ state-building reform

Agents of the Regime? Traditional Leaders and Electoral Politics in South Africa

Daniel de Kadt and Horacio A. Larreguy (2018). *Journal of Politics* 80 (2).

Areas with and without traditional leaders meet the rise of Jacob Zuma as ANC leader; the outcome is the ANC's vote share.

Traditional leaders $\times$ rise of Zuma

- In each study: a **baseline factor** G interacted with an **event** Z that hits every unit

Source: Chen, Wang, and Zhang (2025); de Kadt and Larreguy (2018).

Same DID estimator, a different research design:
“Factorial Difference-in-Differences”

- **Factorial**

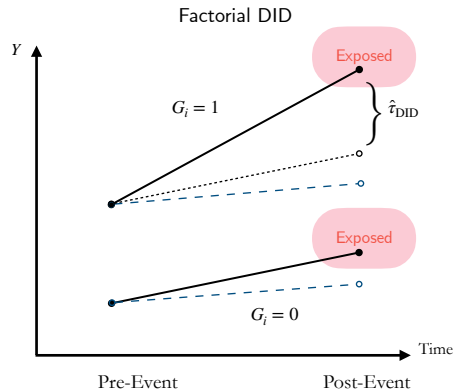
- A classic topic in statistics
- Factorial experiments pioneered by R. A. Fisher and Frank Yates
- ≥ 2 treatment factors: their main effects and interaction are of interest

- **Difference-in-differences**

- Popular in economics and related fields
- A “research design” for causal inference with observational data
- Leverages panel data (units $\times$ time) to identify causal effects, usually the ATT

This Lecture

- Factorial DID is a different research design from a canonical DID
- Under no anticipation & parallel trends, the DID estimator identifies **treatment effect heterogeneity** of exposure to the event (with nuances)
- Identifying G 's **causal effect** requires stronger assumptions
- Factorial DID includes canonical DID as a special case with an additional assumption
- Extensions & practical recommendations



Two groups, two periods; no covariates

- Study population: $i = 1, 2, \dots, n$
- Timing of the event is fixed
- Time periods: $t = \text{pre, post}$
- Baseline factor: $G_i \in \{0, 1\}$
- Data: $\{G_i, Y_{i,\text{pre}}, Y_{i,\text{post}} : i = 1, 2, \dots, n\}$
- **Universal exposure** to the event in the post period:

$$Z_i = 1, \quad i = 1, 2, \dots, n$$

	Pre-Event	Post-Event
$G_i = 1$		$Z_i = 1$
$G_i = 0$		$Z_i = 1$

Factorial DID

	Pre-Event	Post-Event
$G_i = 1$		$Z_i = 1$
$G_i = 0$		$Z_i = 0$

Canonical DID

The Difference-in-Differences Estimator

	Pre-Event	Post-Event
$G_i = 1$	$\frac{1}{n_1} \sum_{i:G_i=1} Y_{i,\text{pre}}$	$\frac{1}{n_1} \sum_{i:G_i=1} Y_{i,\text{post}}$
$G_i = 0$	$\frac{1}{n_0} \sum_{i:G_i=0} Y_{i,\text{pre}}$	$\frac{1}{n_0} \sum_{i:G_i=0} Y_{i,\text{post}}$

Define

- $\hat{\tau}_{\text{DID}} = \frac{1}{n_1} \sum_{i:G_i=1} (Y_{i,\text{post}} - Y_{i,\text{pre}}) - \frac{1}{n_0} \sum_{i:G_i=0} (Y_{i,\text{post}} - Y_{i,\text{pre}})$
- Focus on: $\text{plim } \hat{\tau}_{\text{DID}} = \tau_{\text{DID}} = \mathbb{E}[\Delta Y_i \mid G_i = 1] - \mathbb{E}[\Delta Y_i \mid G_i = 0]$

Potential Outcomes

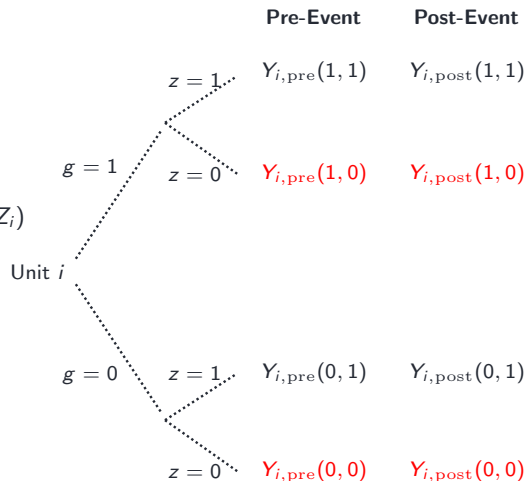
- Frame factorial DID as a factorial design with two factors: g and z

- Potential outcomes indexed by both: $Y_{i,t}(g, z)$

- Observed outcomes:

$$Y_{i,t} = \sum_{g,z \in \{0,1\}} \mathbf{1}\{G_i = g, Z_i = z\} \cdot Y_{i,t}(g, z) = Y_{i,t}(G_i, Z_i)$$

- Recall: in observed data, $Z_i = 1$ for all units $\Rightarrow$ **half of the potential outcomes are never observed**



Red: potential outcomes never observed under universal exposure.

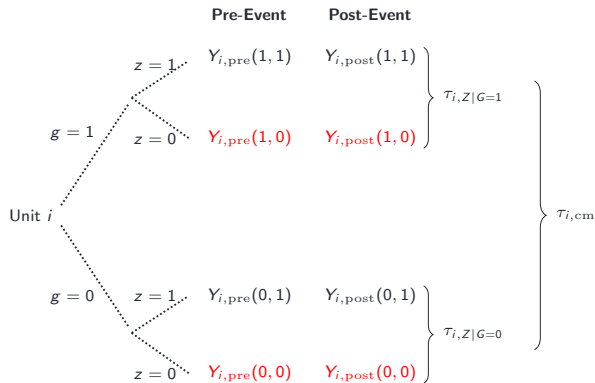
- Individual conditional effect of the event at factor level g :

$$\tau_{i,Z|G=g} = Y_{i,\text{post}}(g, 1) - Y_{i,\text{post}}(g, 0)$$

- Individual interaction effect:

$$\begin{aligned}\tau_{i,\text{cm}} &= Y_{i,\text{post}}(1, 1) - Y_{i,\text{post}}(1, 0) \\ &\quad - Y_{i,\text{post}}(0, 1) + Y_{i,\text{post}}(0, 0) \\ &= \tau_{i,Z|G=1} - \tau_{i,Z|G=0}\end{aligned}$$

the **causal moderation** of G on Z



- **Average causal interaction** (VanderWeele 2009; Bansak 2021):

$$\tau_{\text{cm}} = \mathbb{E}[Y_{i,\text{post}}(1, 1) - Y_{i,\text{post}}(1, 0) - Y_{i,\text{post}}(0, 1) + Y_{i,\text{post}}(0, 0)] = \mathbb{E}[\tau_{i,Z|G=1}] - \mathbb{E}[\tau_{i,Z|G=0}]$$

“Social capital reduced the mortality increase caused by the famine.”

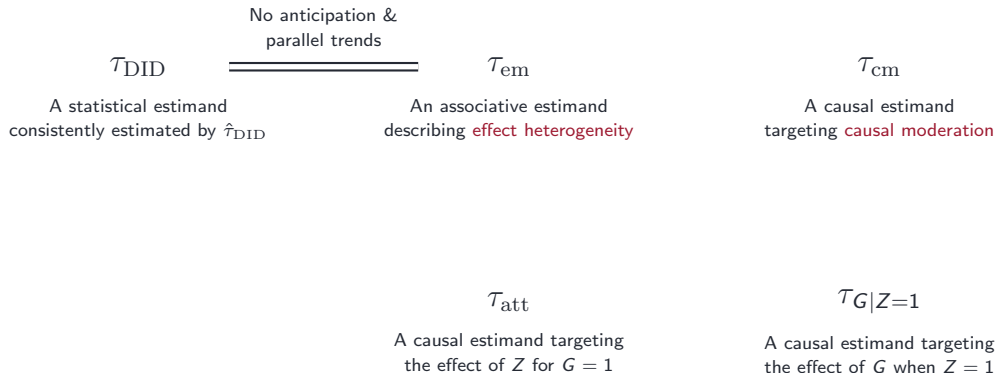
- **Effect modification** — associative:

$$\tau_{\text{em}} = \mathbb{E}[\tau_{i,Z|G=1} \mid G_i = 1] - \mathbb{E}[\tau_{i,Z|G=0} \mid G_i = 0]$$

“Places with higher social capital experienced a smaller mortality increase caused by the famine.”

- The verbs differ because the estimands differ: τ_{cm} is causal, τ_{em} compares **different units**

Identification: A Roadmap



Identification under Canonical DID Assumptions

No Anticipation

$$Y_{i,\text{pre}}(g, 0) = Y_{i,\text{pre}}(g, 1) \text{ for all } i \text{ and } g = 0, 1$$

Parallel Trends

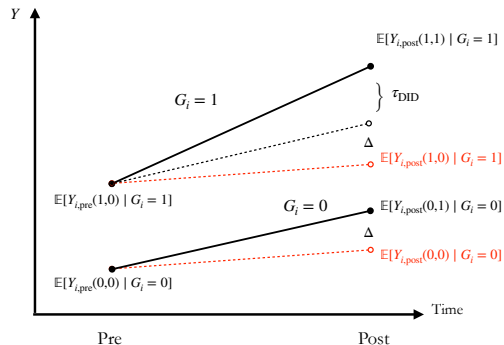
$$\mathbb{E}[\Delta Y_i(1, 0) \mid G_i = 1] = \mathbb{E}[\Delta Y_i(0, 0) \mid G_i = 0]$$

Proposition

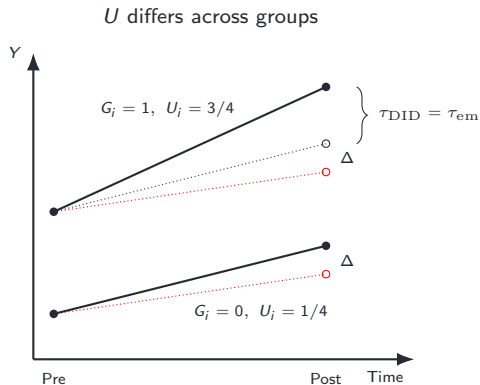
Under no anticipation and parallel trends,

$$\tau_{\text{DID}} = \tau_{\text{em}} = \mathbb{E}[\tau_{i,Z|G=1} \mid G_i = 1] - \mathbb{E}[\tau_{i,Z|G=0} \mid G_i = 0]$$

Two different **conditional effects**, across two different groups



Why τ_{DID} May Not Identify Causal Moderation under Parallel Trends



- Imagine an unobservable U that determines how units respond to the event
 - If both groups share the same U , the event moves them equally and $\tau_{\text{DID}} = 0$, whether U is small or large
- U may be correlated with G : here, high- U units ($U_i = 3/4$) are concentrated in $G = 1$
- Then $\tau_{\text{DID}} = \tau_{\text{em}}$ mixes the moderation of G with compositional differences in U
- Therefore τ_{DID} cannot be interpreted as the causal moderation of G

Identifying Causal Moderation

- Under no anticipation and parallel trends:

$$\tau_{\text{DID}} = \tau_{\text{em}} = \mathbb{E}[Y_{i,\text{post}}(1, 1) - Y_{i,\text{post}}(1, 0) \mid G_i = 1] - \mathbb{E}[Y_{i,\text{post}}(0, 1) - Y_{i,\text{post}}(0, 0) \mid G_i = 0]$$

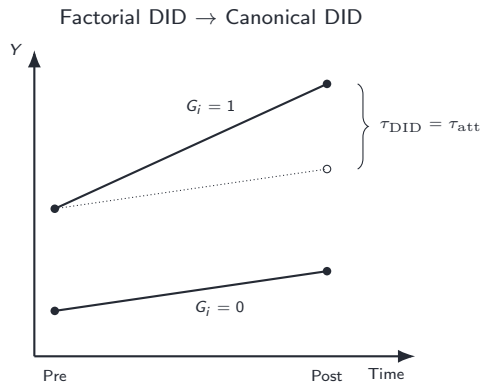
- When does τ_{em} equal τ_{cm} , which drops the conditioning on G_i ?

Factorial Parallel Trends (sufficient condition)

$\Delta Y_i(g, z) \perp\!\!\!\perp G_i$ (orthogonality between G and first-differenced potential outcomes)

- Weaker than (quasi-)random assignment of G : $Y_{i,t}(g, z) \perp\!\!\!\perp G_i$
- Under no anticipation, parallel trends, and $\Delta Y_i(g, z) \perp\!\!\!\perp G_i$: $\tau_{\text{DID}} = \tau_{\text{em}} = \tau_{\text{cm}}$

Reconciling Canonical and Factorial DID



Exclusion Restriction on Z for Group $G_i = 0$

$$\mathbb{E}[Y_{i,\text{post}}(0, 1) \mid G_i = 0] = \mathbb{E}[Y_{i,\text{post}}(0, 0) \mid G_i = 0]$$

the event does not move $G = 0$ units: their Δ is zero

Proposition

Under no anticipation, parallel trends, and the exclusion restriction,

$$\tau_{DID} = \tau_{att} = \mathbb{E}[Y_{i,\text{post}}(1, 1) - Y_{i,\text{post}}(1, 0) \mid G_i = 1]$$

- This is exactly the canonical DID of Lecture 2: “untreated” units are units for which the event is assumed not to matter

Identifying G 's Conditional Effect

- One more question the interaction might answer: the **effect of G itself** during the event

Exclusion Restriction on G Absent the Event

$$\mathbb{E}[Y_{i,\text{post}}(1, 0)] = \mathbb{E}[Y_{i,\text{post}}(0, 0)]$$

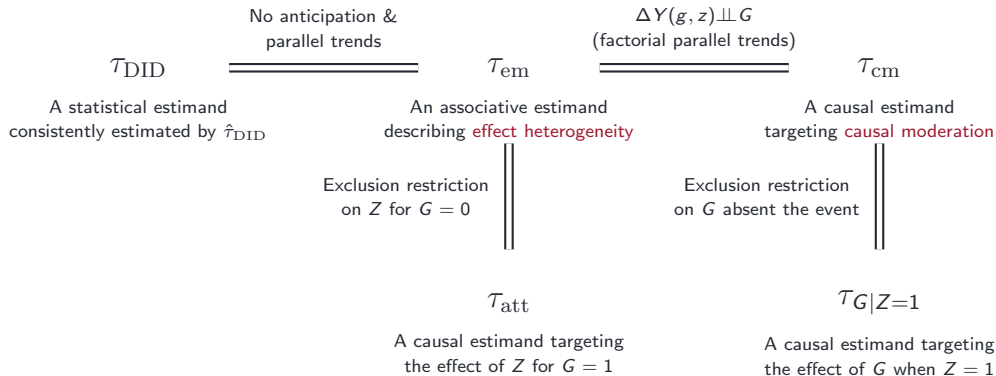
without the event, the baseline factor would not move the outcome

- Under no anticipation, parallel trends, factorial parallel trends, and this exclusion restriction:

$$\tau_{\text{DID}} = \tau_{\text{cm}} = \mathbb{E}[Y_{i,\text{post}}(1, 1) - Y_{i,\text{post}}(0, 1)] = \tau_{G|Z=1}$$

- In the famine example: the average causal effect of social capital on mortality **during the famine years**

Summary of Identification Results



Canonical vs Factorial DID: Two Research Designs

Canonical DID (reframed)

Define the canonical DID research design as the combination of:

- 2×2 data structure & **universal exposure**
- Identification: under no anticipation & parallel trends & **exclusion restriction on Z** , $\hat{\tau}_{\text{DID}}$ identifies τ_{att}

Factorial DID

Define the factorial DID research design as the combination of:

- 2×2 data structure & **universal exposure**
- Identification:
 - ① Under no anticipation & parallel trends, $\hat{\tau}_{\text{DID}}$ identifies τ_{em}
 - ② Adding $\Delta Y_i(g, z) \perp\!\!\!\perp G_i$, $\hat{\tau}_{\text{DID}}$ identifies τ_{cm}
 - ③ Adding also the **exclusion restriction on G** , $\hat{\tau}_{\text{DID}}$ identifies $\tau_{G|Z=1}$

Conditioning on Covariates in Practice

- Researchers often run the following TWFE regression:

$$Y_{it} = \alpha_i + \xi_t + \tau G_i \cdot \text{Post}_t + \beta^T X_i \cdot \text{Post}_t + \epsilon_{it}$$

Table 1

Clans and mortality rate during the great famine.

Cao, Xu & Zhang (2022)

Outcome variable mean	Outcome variable: Mortality rate (%)					
	(1)	(2)	(3)	(4)	(5)	(6)
	13.997	13.997	13.997	13.997	13.997	13.997
High clan density x Famine period	-1.456*** (0.596)	-2.325*** (0.655)	-2.254*** (0.698)			
Log (#Genealogies/pop) x Famine period				-3.385*** (0.398)	-4.108*** (0.597)	-4.296*** (0.661)
Grain output (PC) x Famine period		0.006* (0.003)	0.005* (0.003)		0.005* (0.003)	0.005* (0.003)
Non-farming land ratio x Famine period		0.006 (0.031)	0.002 (0.033)		0.013 (0.031)	0.009 (0.033)
Urbanization rate x Famine period		-0.075** (0.031)	-0.076** (0.034)		-0.088** (0.030)	-0.087** (0.033)
Distance from Beijing x Famine period		0.004*** (0.001)	0.004*** (0.001)		0.004*** (0.001)	0.004*** (0.001)
Distance from provincial capital x Famine period		0.002 (0.003)	0.003 (0.003)		0.001 (0.003)	0.002 (0.003)
Historical migrants * Famine period		-0.078*** (0.011)	-0.081*** (0.012)		-0.053*** (0.011)	-0.055*** (0.013)
Crop suitability index for rice * Famine period		0.359*** (0.160)	0.392*** (0.171)		0.460*** (0.164)	0.470*** (0.175)
Share of minority * Famine period		-0.049*** (0.017)	-0.050*** (0.018)		-0.067*** (0.016)	-0.067*** (0.018)
Averaged years of schooling * Famine period		-0.822** (0.406)	-0.899** (0.431)		-0.505 (0.408)	-0.372 (0.434)
Observations	17,342	17,342	17,342	17,342	17,342	17,342
Number of counties	1448	1448	1448	1448	1448	1448
R-squared	0.388	0.407	0.428	0.390	0.409	0.430
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
County-specific time trends	No	No	Yes	No	No	Yes

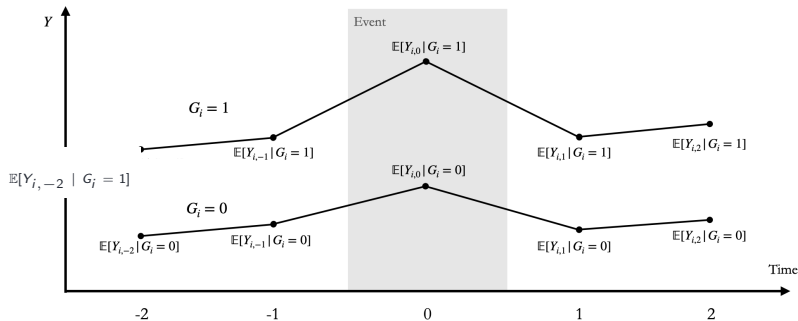
Notes: In columns (1)-(3), high clan density is a dummy variable indicating whether the number of genealogies in a county divided by the county's population size is above the national mean. In columns (4)-(6), we use the continuous measure. Famine period is defined 1958-1961. Grain output per capita, non-farming land ratio, and urbanization rate are measured in 1957; distance from Beijing and distance from provincial capital are spherical distances; crop suitability index for rice is from GAEZ (Global Agro-Ecological Zones); historical migrants denote the inflow of prominent families in the late Song Dynasty; the share of ethnic minorities and averaged years of schools are calculated based on people older than 40 years old in 1982 using the 1982 population census data. See Appendix B for details on data source. Standard errors clustered at the county level are reported in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1. For easier comparisons of the coefficients, all regressions

- Believe $\Delta Y_i(g, z) \perp\!\!\!\perp G_i \mid X_i$ is more plausible than $\Delta Y_i(g, z) \perp\!\!\!\perp G_i$
- Simple improvements to the TWFE regression: (a) demean X_i ; (b) add interaction terms

$$Y_{it} = \alpha_i + \xi_t + \tau G_i \cdot \text{Post}_t + \beta^T X_i \cdot \text{Post}_t + \gamma^T G_i \cdot X_i \cdot \text{Post}_t + \epsilon_{it}$$

- $\hat{\tau}$ identifies $\mathbb{E}[\tau_{\text{em}}(x)]$, or τ_{cm} under linearity
- Requires overlap: $0 < \mathbb{P}(G_i = 1 \mid X_i = x) < 1$
- Can leverage more flexible models of ΔY_i on X_i for subgroups $G_i = 1, 0$:
 - Transform data into wide form; replace Y_i with ΔY_i
 - Apply estimators developed for selection-on-observables designs (stratification, matching, balancing, IPW, AIPW, outcome modeling, double machine learning, ...)

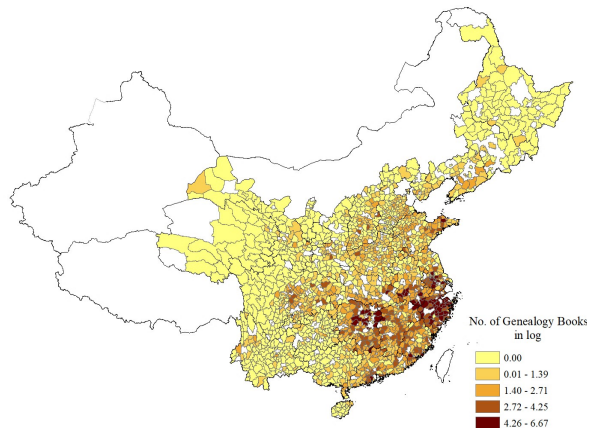
Extension to Multiple Pre- and Post-Periods



- Lesson 1: pretrend tests can help assess parallel trends, but **not** $\Delta Y(g, z) \perp\!\!\!\perp G$
- Lesson 2: using post-periods as non-event periods requires an additional “no carryover effect” assumption

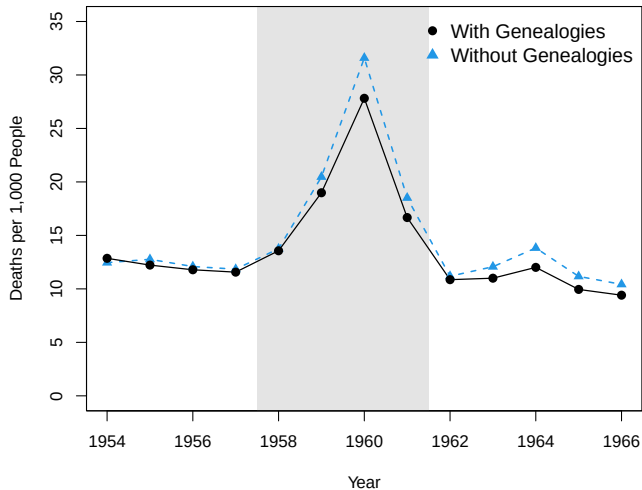
Example: Clans and Calamity

- Event: the Great Famine (1958–1961)
- G: social capital, proxied by genealogy books
 - No genealogies: 412 counties
 - Have pre-PRC genealogies: 509 counties
- Outcome: county mortality rate



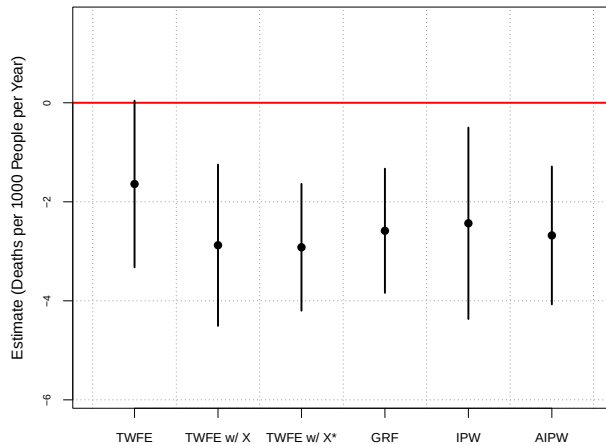
Source: Cao, Xu, and Zhang (2022); replication data in the `fdid` R package.

Raw Data: Group Means



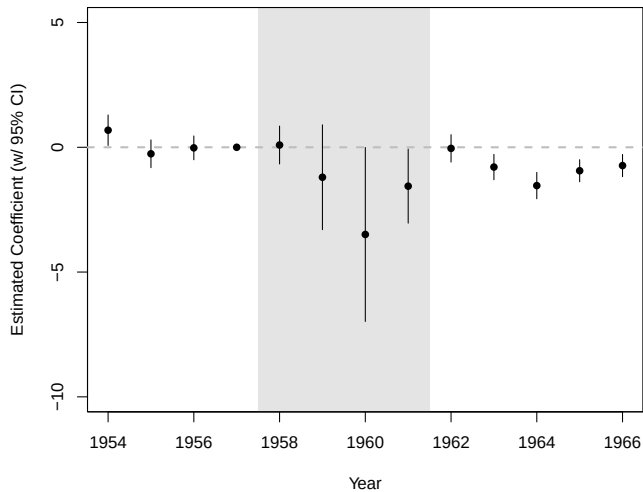
Conditional on Covariates

- Pre-event covariates:
 - Log population size
 - Per capita grain production
 - Ratio of non-farming land
 - Urbanization ratio
 - Distance from Beijing
 - Distance from the provincial capital
 - Share of ethnic minorities
 - Rice suitability
 - Average years of education



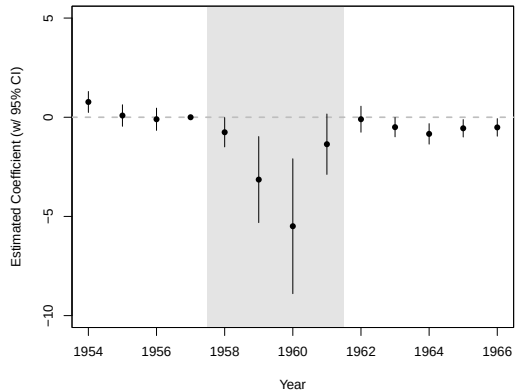
- Estimates are stable across estimators, from TWFE to AIPW

Dynamic Estimates: DID

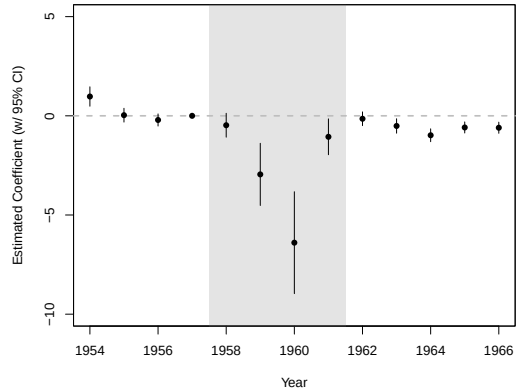


Dynamic Estimates: TWFE with Covariates

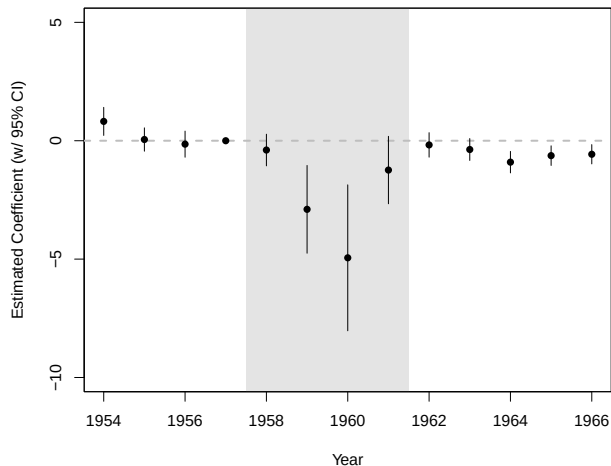
TWFE⁺: covariates $\times$ Post



TWFE*: additional interaction terms

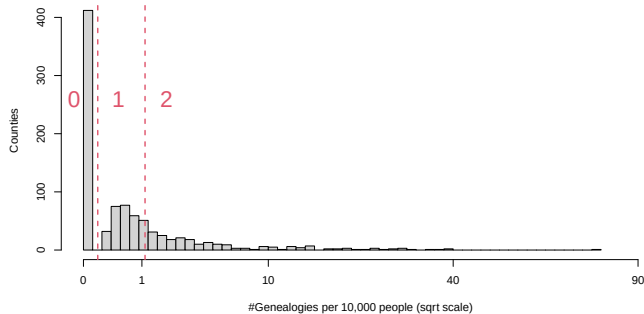


Dynamic Estimates: AIPW



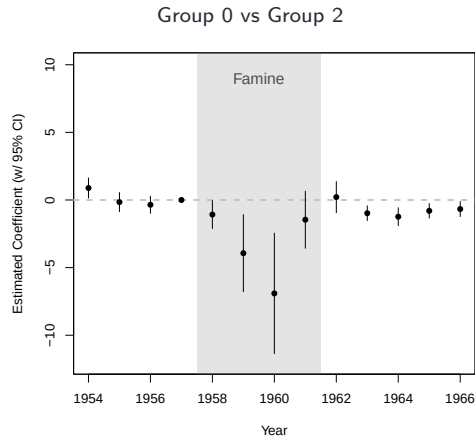
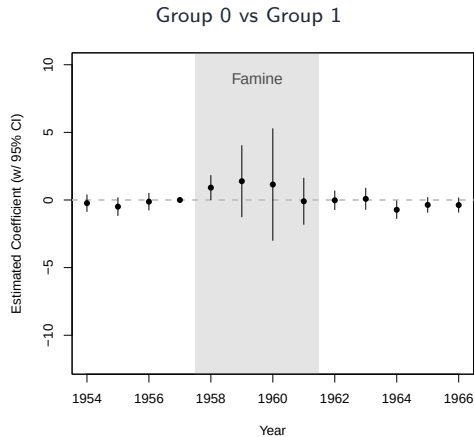
- No visible pretrend; the mortality gap opens exactly in the famine years

Multi-valued G



- $G = 0$: 412 counties (no genealogies)
- $G = 1$: 254 counties (some genealogies)
- $G = 2$: 255 counties (many genealogies)

Multi-valued G: AIPW Estimates



- The mortality gap concentrates where social capital is strongest

How to Correctly Interpret Your Findings?

Estimand	Interpretation in the running example	Key identifying assumptions
τ_{em} , effect modification	How did the average effect of exposure to famine on mortality differ between counties with high and low social capital?	No anticipation; parallel trends
τ_{att} , effect of exposure for $G = 1$	What was the average effect of exposure to famine on mortality in counties with high social capital?	+ exclusion restriction on Z in group $G = 0$
τ_{cm} , causal moderation	To what extent did high (vs. low) social capital mitigate the impact of famine on mortality?	No anticipation; parallel trends; $\Delta Y(g, z) \perp\!\!\!\perp G \mid X$
$\tau_{G Z=1}$, G 's conditional effect	What was the average causal effect of social capital on mortality during the famine years?	+ exclusion restriction on G absent the event

```
library(fdid)
mortality$any_gen <- as.integer(
  mortality$pczupu > 0) # 412 vs 509 counties
s <- fdid_prepare(
  data = mortality,
  Y_label = "mortality",
  G_label = "any_gen",
  unit_label = "countyid",
  time_label = "year")

fit <- fdid(s, tr_period = 1958:1961,
  ref_period = 1957,
  method = "ols1")
```

Observed interaction

The famine $\times$ social-capital coefficient is an effect-modification contrast.

Moderation interpretation

Causal moderation requires factorial parallel trends.

- What is your estimand?! (Lundberg, Johnson, and Stewart 2021)
- Assess no anticipation and parallel trends (e.g., by checking pretrends)
- If the goal is causal moderation, justify $\Delta Y(g, z) \perp\!\!\!\perp G \mid X$ and show robustness of findings (e.g., by conducting sensitivity analysis)
- If using TWFE models, demean covariates and add interaction terms $G_i \cdot X_i \cdot \text{Post}_t$; check overlap
- Should not automatically assume no carryover effects

- When an event hits every unit, there is no untreated group; FDID compares before-after changes across levels of a **baseline factor**
- The arithmetic is canonical DID; the estimand is not: under no anticipation and parallel trends, τ_{DID} is **effect modification** — an associative quantity
- **Factorial parallel trends** ($\Delta Y(g, z) \perp\!\!\!\perp G$) upgrades it to **causal moderation**; an exclusion restriction on G upgrades it to G 's conditional effect; an exclusion restriction on Z recovers canonical DID
- The broader lesson of these lectures: **estimators are not research designs** — the same arithmetic answers different questions under different assumptions
- Lecture 4: back to TWFE regressions, and what breaks under staggered timing

- `fdid` (R): factorial DID with the famine application data
- `panelView` (R & Stata): treatment status and outcome visualization
- Tutorials: <https://yiqingxu.org/tutorials/panel.html>

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